EPPS6323 Knowledge Mining

Assignment 1 – Brush up R and Quarto

1. Review/refresh R programming:
   a. Resources:
      i. Introduction to R (University of Southern California)
      ii. EPPS Math and Coding Camp (Start from Day 3)
2. Lab01, Lab02 in R (Download the .qmd files in Teams)
   a. Render and upload your files to GitHub website
   b. Be sure to uploaded associated folders (e.g. Lab01.html, Lab01_files, Lab02.html, Lab02_files)
3. Review Chapters 3 to 7 in R4DS, Section 3.5 in Data programming. Run an exploratory data analysis with R using the TEDS2016 dataset (sample codes as follows):

```
library(haven)
TEDS_2016 <-
haven::read_dta("https://github.com/datageneration/home/blob/master/DataProgra
mming/data/TEDS_2016.dta?raw=true")
```

4. What problems do you encounter when working with the dataset?
5. How to deal with missing values?
6. (Next step) Explore the relationship between `Tondu` and other variables including `female, DPP, age, income, edu, Taiwanese` and `Econ_worse`. What methods would you use?
7. (Next step) How about the `votetsai` variable (vote for DPP candidate Tsai Ing-wen)?
8. (Next step) Generate frequency table and barchart of the `Tondu` variable. Assign labels to the variable using the following:

```
TEDS_2016$Tondu<-as.numeric(TEDS_2016$Tondu,labels=c("Unification now”,
“Status quo, unif. in future”, “Status quo, decide later", "Status quo
forever", "Status quo, indep. in future", "Independence now”, “No response"))
```